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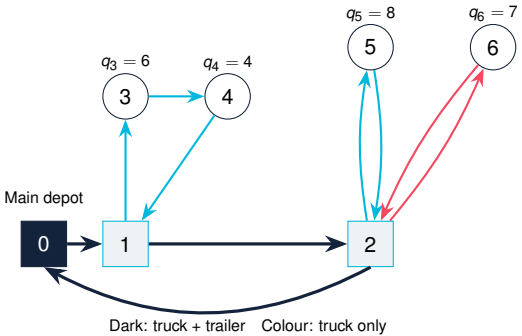
Truck and trailer routing with satellite depots

STTRPSD — Metaheuristics course project

Three sessions

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One vehicle and two kinds of trip



Truck capacity $Q_V = 10$: the trips carry 10, 8, and 7.

The vehicle's journey

- Truck and trailer visit selected satellites.
- The trailer stays parked while the truck serves customers.
- Each truck trip returns to its starting satellite.
- Every customer is served exactly once.

The optimization problem

Minimize the complete distance

$$\min \quad \underbrace{\text{main-tour distance}}_{\text{truck + trailer}} + \underbrace{\text{sum of truck-trip distances}}_{\text{truck only}}$$

Choose the satellites and their order, assign customers to trips, and order the customers within each trip.

- Each truck trip carries at most Q_V.
- The main tour starts and ends at depot 0, visiting each chosen satellite once.
- Distances are Euclidean, **without integer rounding**.
- The scorer checks feasibility and recomputes the full distance.

Villegas et al. (2010), *Engineering Applications of Artificial Intelligence* 23, 780–794. https://doi.org/10.1016/j.engappai.2010.01.013

Greedy construction, local search, then a metaheuristic

1. Greedy construction

Implement a rule that constructs feasible routes. Validate the constructor on the ten practice instances and record a baseline.

2. Local search

Improve routes with relocation, swap, 2-opt, or other justified moves.
Compare the results with the greedy constructor.

3. Metaheuristic

Implement a genetic, population-based, swarm, or other justified metaheuristic. Compare all three stages under the **30-second budget**.

Reuse your constructor and local-search operators. The paper you pitched can guide the final method.

Project focus

Design and evaluate a metaheuristic. The statement's integer model is an optional aid to understanding the constraints.

The solver interface and benchmark

A common interface

```
solve --instance instance.json
      --solution solution.json
      --seed 0 --time-limit 30
```

Solvers can use any programming language.

A main tour and rooted truck trips

$$M = (0, 1, 2, 0),$$
$$r_1 = (1; 3, 4),$$
$$r_2 = (2; 5), \qquad r_3 = (2; 6).$$

Trip format: (satellite; customers).

- **10 practice instances** for every team.
- **10 other assessment instances** with similar characteristics.
- **30 seconds per run**, including startup.
- Three common seeds and one CPU core.
- Failed or invalid runs earn zero for that run.

Five-minute exchanges with every group

Exchange	S1	S2	S3
11:30–11:35	G4	G5	G6
11:36–11:41	G1	G3	G7
11:42–11:47	G2	G7	G5
11:48–11:53	G7	G6	G2
11:54–11:59	G6	G4	G3
12:00–12:05	G3	G1	G4
12:06–12:11	G5	G2	G1

During the exchange

Every member answers.
No computers, phones, or notes.

One minute between groups.
Other groups continue working.

Including the paper pitch, every group has mean position 4. No group repeats an oral position.

The final mark out of 20

Component	Raw	Final
Participation P	/20	/10
Code C	/4	/2
Report R	/8	/4
Benchmark performance K	/8	/4

Combine the raw marks

$$G = \frac{P + C + R + K}{2}$$

Participation

Paper pitch and three exchanges: **four marks out of 5**.

The report has five sections

1. **Method description**: representation, construction, search, and stopping rule.
2. **Benchmarking and results**: protocol, repeated runs, tables, and figures.
3. **Analysis and conclusions**: explain the results, tradeoffs, and limitations.
4. **Instructions to run the programme**: dependencies and exact commands.
5. **One page on generative AI**: your interactions, checks, and each student's contribution.

Session 1: a greedy solver

1. Score the supplied toy solution and account for every cost term.
2. Choose a greedy decision rule and implement your constructor.
3. Validate all ten practice instances and record the baseline results.
4. Explain each decision and identify a neighbourhood for Session 2.

Be ready to explain your method

Every member should be able to explain the greedy rule and why the resulting routes are feasible.